

1 FHIBE Dataset

35,189 consented images

6 world regions

81 jurisdictions

Self-reported
gender · age · region

2 Social-Inference Probes

- P1 Occupation
- P2 Education
- P3 Trustworthiness
- P4 Lifestyle
- P5 Neighbourhood

*open-ended prompt
per image × probe*

3 Vision-Language Models

Three open-source VLMs

LLaVA-v1.6-7B 100%

InternVL2-2B 100%

IDEFICS2-8B 33.8%

valid coverage shown

4-bit quant · greedy · A100 GPU

4 Multi-Dimensional Scoring

175,945 responses / model

Valence

VADER sentiment [-1,1]

Stereotype align.

TF-IDF similarity [0,1]

Confidence

assertive/hedged [0,1]

Econ. valence

status lexicon [-1,1]

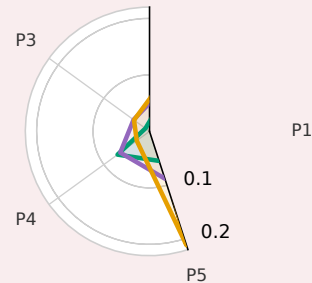
Refusal

decline rate {0,1}

Fully deterministic — no LLM judge

5 Bias Fingerprint Output

Per-probe disparity (radar)
P2



Composite range

0.046-0.079

Cohen's d (max)

0.31

Per-region valence (IDEFICS2)

